

Green Fleet Optimizer — Results for the Idea Submission

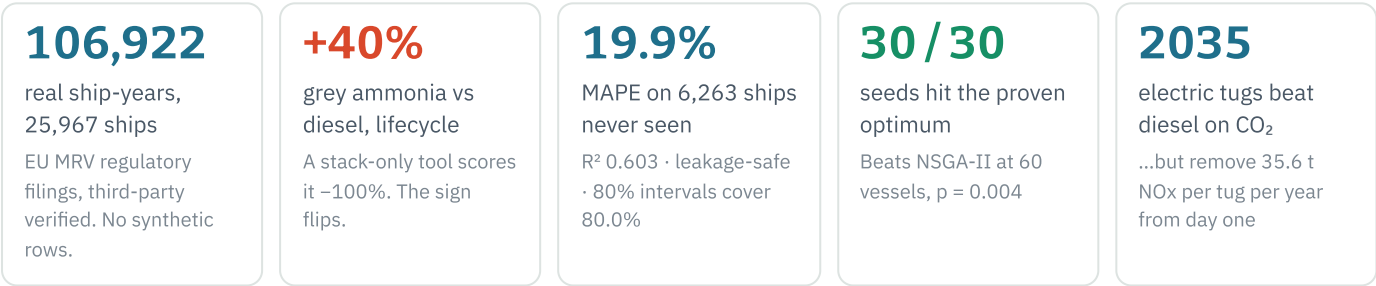
Quantum-inspired fuel consumption prediction and green fleet optimization. Every figure on this page is read from `artifacts/*.json` or recomputed with a fixed seed — nothing is typed from memory, so the deck cannot drift from the code.

Data: EU MRV 2018–2025, 106,922 ship-years, real and audited **Tests:** 399 passing

Charts: 14, at 300 dpi, sized for 16:9

IF A JUDGE READS NOTHING ELSE

Five numbers to lead with



The fifth number wins the room. It is specific, uncomfortable, and obviously not sales copy — which makes every other claim on this page more credible to a government audience.

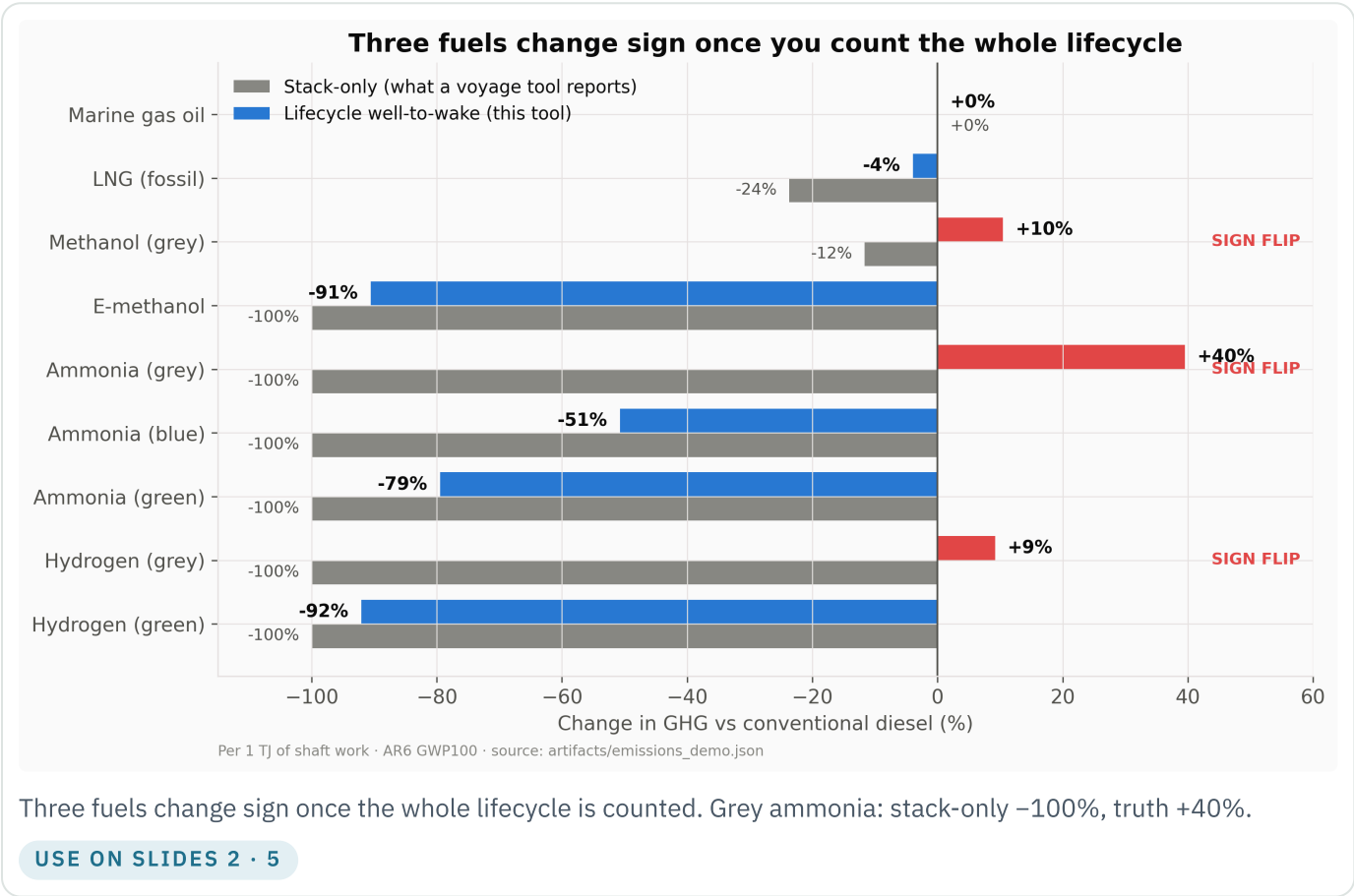
Title

FIELD	VALUE
Problem Statement ID	SIH26138
Problem Statement Title	Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization
Organisation	Egreen Quanta
Theme	Transportation & Logistics (confirm on portal)
PS Category	Software
Tagline	India's first sovereign, quantum-ready planner that decides which vessels, which fuels, and which speeds — not just which route.

Proposed solution

The problem, in our words. Ports and coastal operators must decide which ships to buy, which fuels to commit to, how fast to sail, and when — under cargo demand, schedules, emission caps, and fuel that is only bunkerable at some Indian ports in some years. Twenty ships, four fuels and five speed bands is $\sim 10^{20}$ combinations. ZeroNorth, Wärtsilä FOS and DeepSea optimise *one voyage of one ship* and count only funnel CO₂. Nobody decides the fleet.

The solution, in one sentence. A decision-support system with two engines: a physics-informed fuel predictor trained on 26,000 real ships, and a quantum-inspired fleet optimizer that outputs a Pareto front of feasible plans under well-to-wake lifecycle emissions and Indian port constraints.



What is unique — four things nobody else has

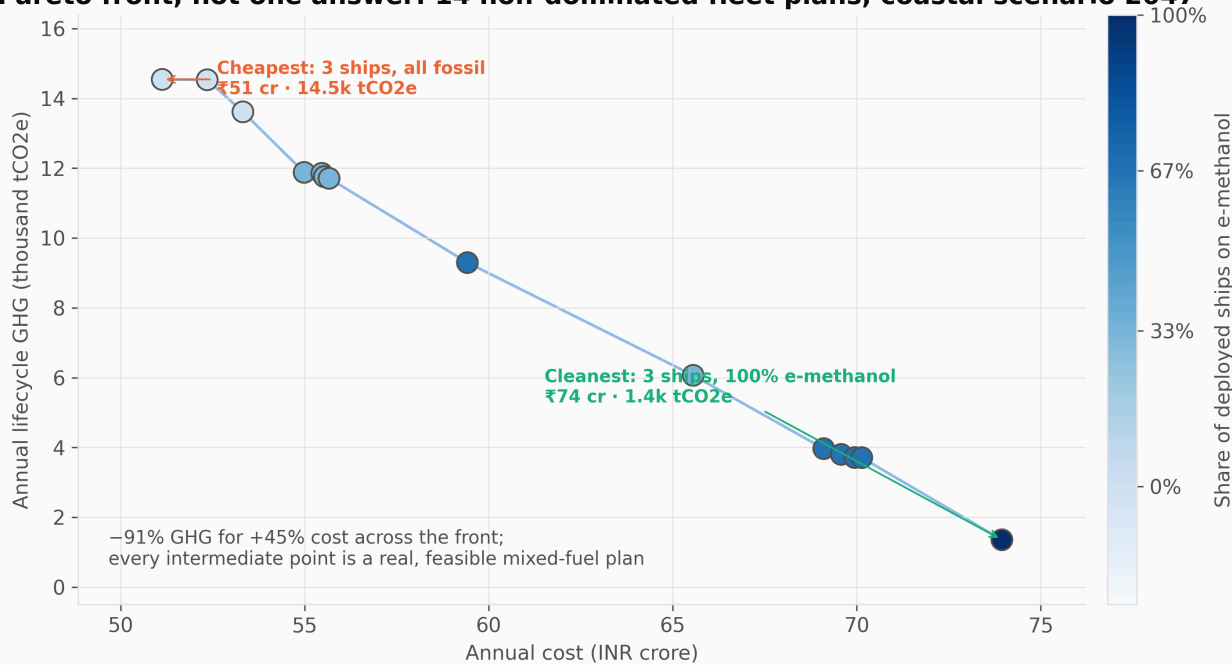
UNIQUENESS	PROOF POINT	CHART
Lifecycle, not funnel — CH ₄ slip, N ₂ O, pilot fuel, grey/blue/green pathways	Grey ammonia +40% · grey methanol +10% · grey hydrogen +9% — all scored as improvements by a stack-only model	01
Measured, not defaulted — slip from 745 measured LNG carrier ship-years	Default 3.1% says LNG is +5.5% <i>worse</i> ; measured 1.72% says -3.9% <i>better</i> . They disagree on the sign.	02
India-specific — 8 ports, year-indexed bunkering, CEA grid, Harit Sagar, GTTP, Mormugao	Electric tug: +20% CO ₂ in 2030, -4% in 2035, crossover 2035 — while removing 35.6 t NOx/yr immediately	03 · 04
Quantum-inspired and quantum-ready — Q-bit + rotation gates, QPSO, QUBO	30/30 exact vs proven MILP · beats NSGA-II at 60 vessels, p = 0.004 · QUBO ground state matches CP-SAT exactly	09 · 11

Against the competition

CAPABILITY	ZERONORTH / WÄRTSILÄ / DEEPSEA	THIS TOOL
Decision level	one voyage, one vessel	fleet mix, fuel, speed, over years
Emissions basis	tank-to-wake CO ₂	well-to-wake CO ₂ e: CH ₄ + N ₂ O + pilot + upstream
Methane slip	a default, if at all	measured from 745 ship-years
Grey vs green pathways	not modelled	modelled; grey ammonia comes out worse than diesel
Local air pollutants	not reported	NOx / SOx / PM tracked separately
Indian port fuel availability	not modelled	year-indexed, 8 major ports
Indian grid carbon	not modelled	CEA factor + T&D losses + decarbonisation path
Harit Sagar / GTTP	none	KPI glide path, on/off-track
Quantum-readiness	none	QUBO formulation, verified
Data residency	foreign SaaS	on-premises, offline-capable

Say it this way: "Those are strong products doing a different job. They optimise a voyage. Nothing in them decides which vessels to buy and which fuels to commit to – which is the decision a port under GTTP and Harit Sagar actually faces."

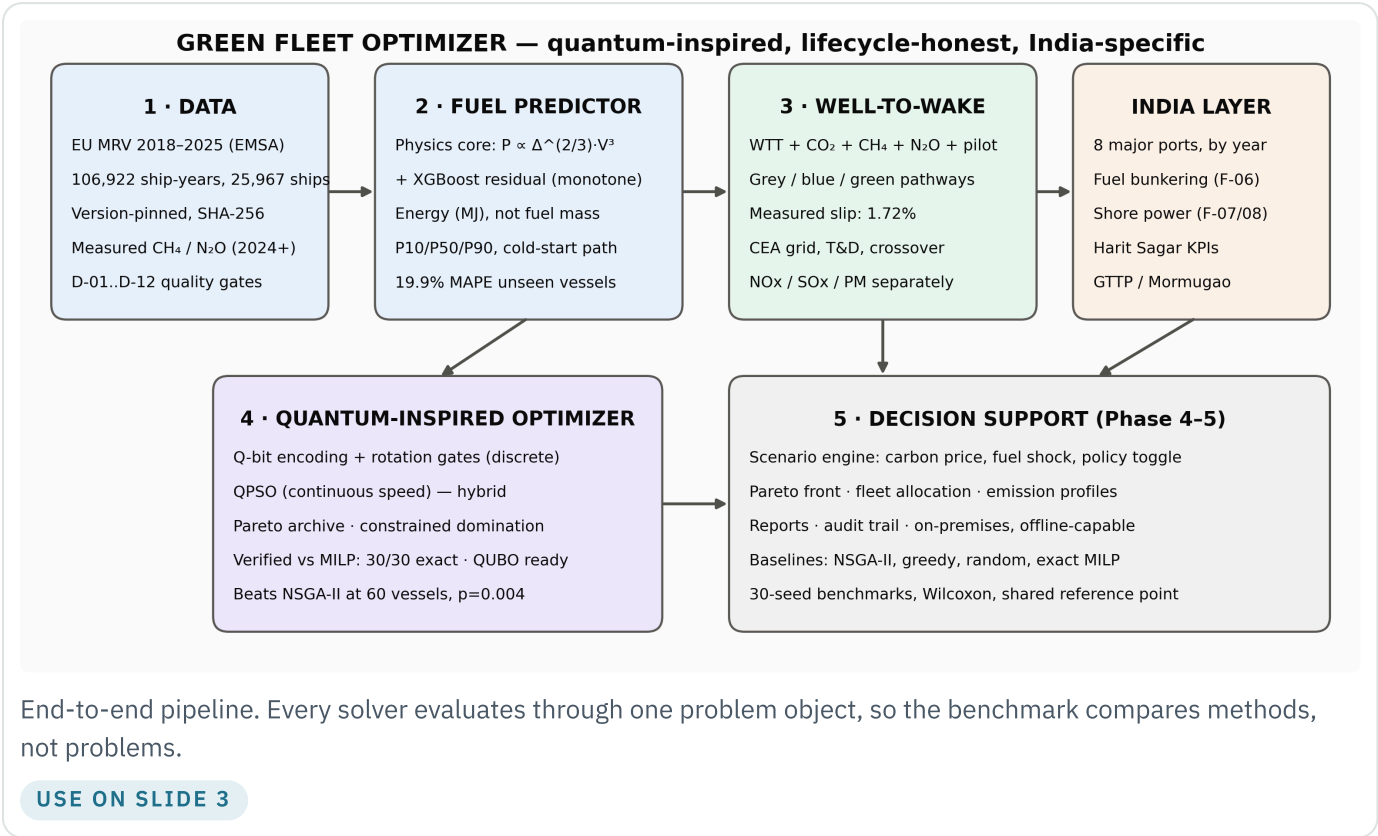
A Pareto front, not one answer: 14 non-dominated fleet plans, coastal scenario 2047



12 candidate vessels · 3 Indian coastal routes · year 2047 · MGO/HFO/LNG + e-methanol · every point meets all cargo demand and every constraint

A Pareto front, not one answer: –91% GHG for +45% cost, with 14 feasible mixed-fuel plans between. Coastal scenario, 2047.

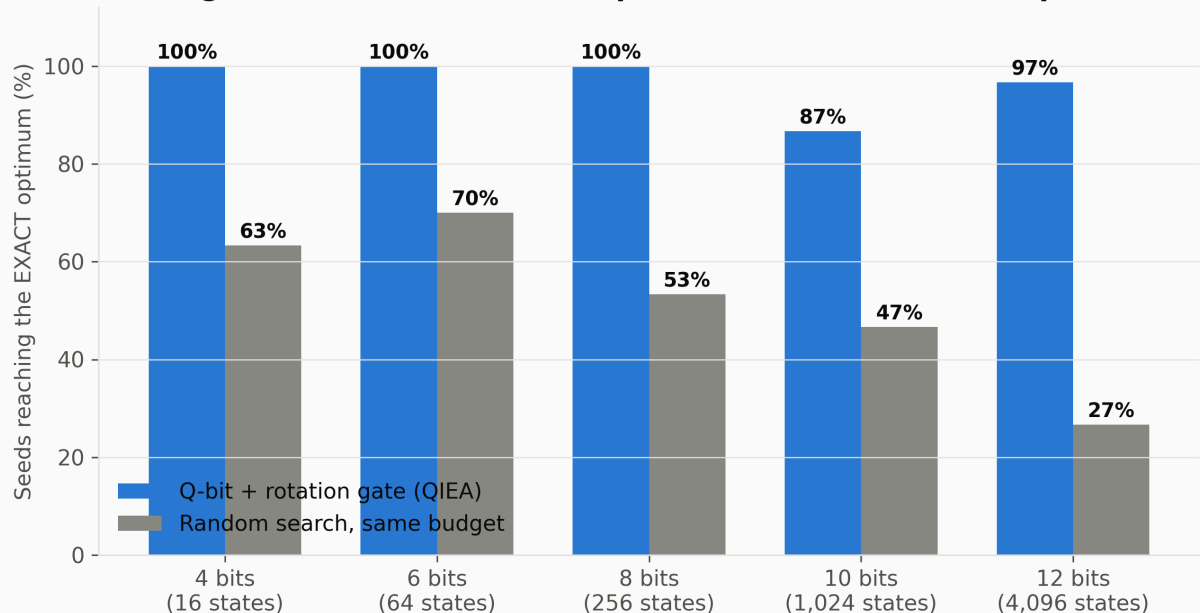
Technical approach



The rotation angle — judges ask

$\delta = 0.03\pi \cdot n_{\text{bits}} / n_{\text{generations}}$, scaled to problem size and budget and calibrated by sweep, not inherited. A fixed angle tuned on a 14-bit problem lost two orders of magnitude of hypervolume on a 360-bit fleet; the size-scaled rule satisfies the 2-, 12- and 60-vessel instances with one setting.

Verified against brute force: exact optimum in 100% of seeds up to 2^8 states

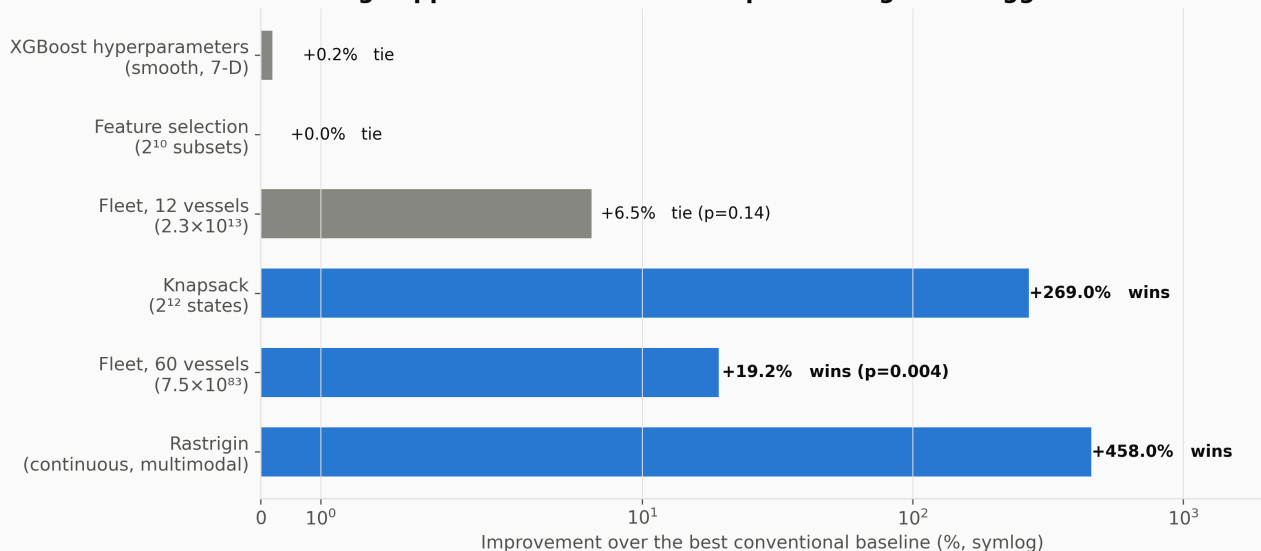


Knapsack with a known exhaustive optimum · 30 seeds · budget matched per seed (B-01, Q-09)

Verified against brute force: exact optimum in 100% of seeds up to 2^8 states.

SLIDES 3 · 4

The advantage appears when the search space is large and rugged - and not otherwise

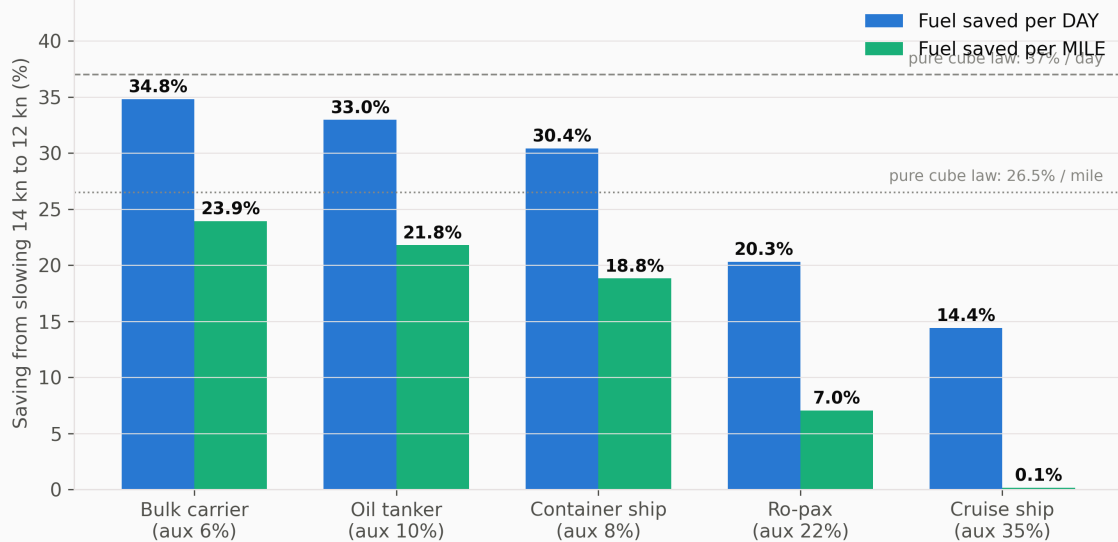


Every comparison budget-matched. Ties are reported as ties. Knapsack/Rastrigin: gain in exact-hit rate / objective vs random search.

Where quantum-inspired search pays off — and where it ties. Reported either way.

SLIDES 3 · 4

Slow steaming pays off very differently by vessel: a cruise ship saves almost nothing per mile



Separating propulsion (V^3) from speed-independent hotel load is what a universal cube law cannot express.

Slow steaming saves 23.9%/mile on a bulk carrier and 0.1%/mile on a cruise ship — a universal cube law cannot express this.

SLIDES 3 · 5

Tech stack. Python 3.10 · NumPy · pandas · scikit-learn · XGBoost (monotone constraints) · pymoo (NSGA-II) · OR-Tools CP-SAT (exact MILP) · SciPy (Wilcoxon) · own QIEA / QPSO / QUBO in NumPy · pytest (399 tests) · ruff · uv. Planned: FastAPI, React + Plotly, PostgreSQL, Docker — on-premises, offline.

Feasibility & viability

19.9%

MAPE, unseen vessels
medAPE 13.4% · R² 0.603

24.4%

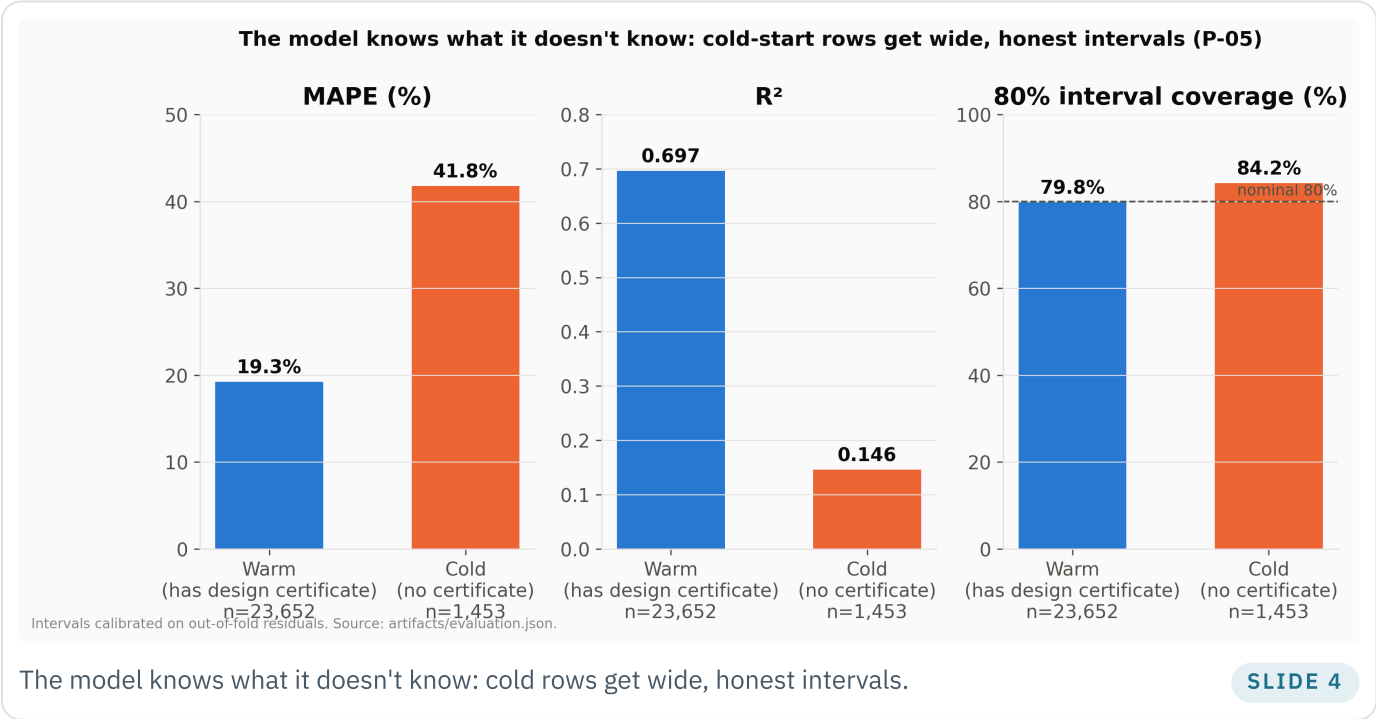
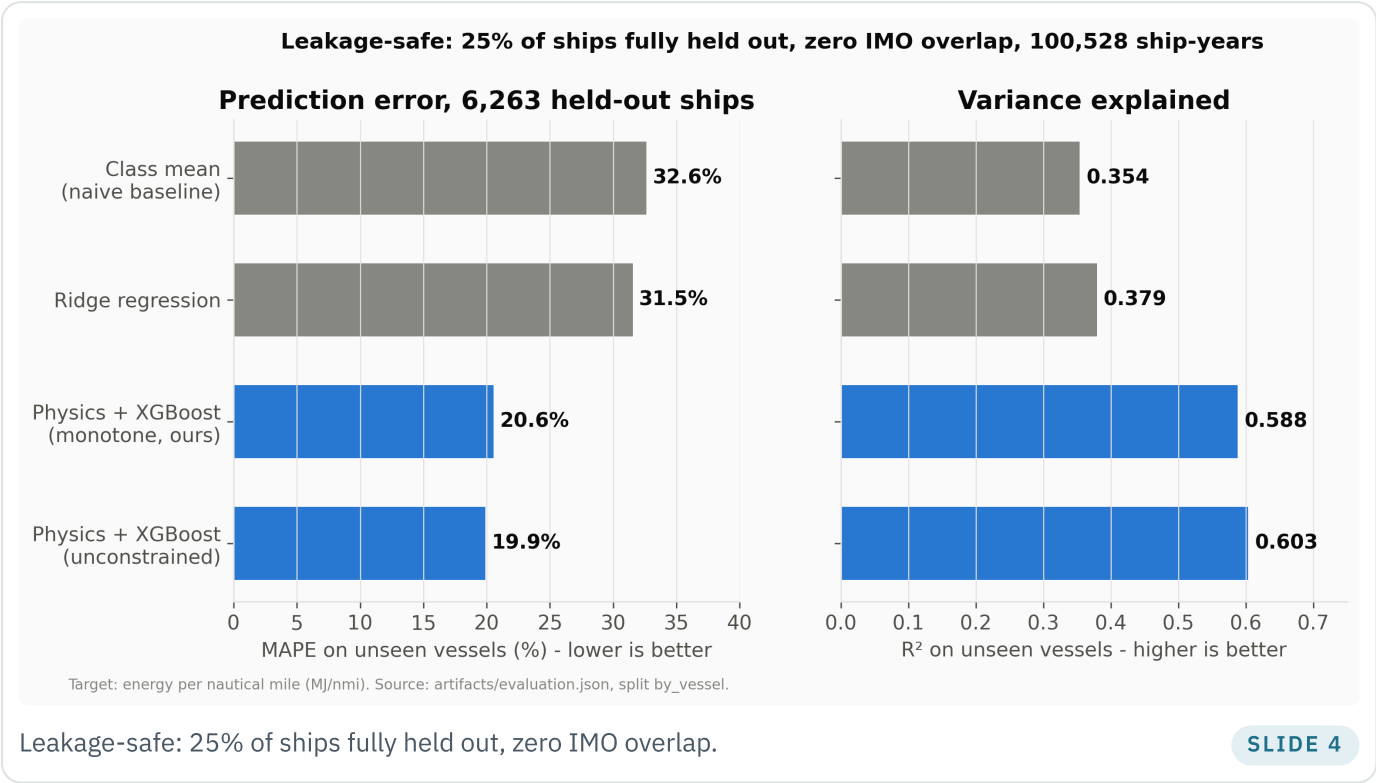
MAPE, unseen years 2024–25
the strictest split

80.0%

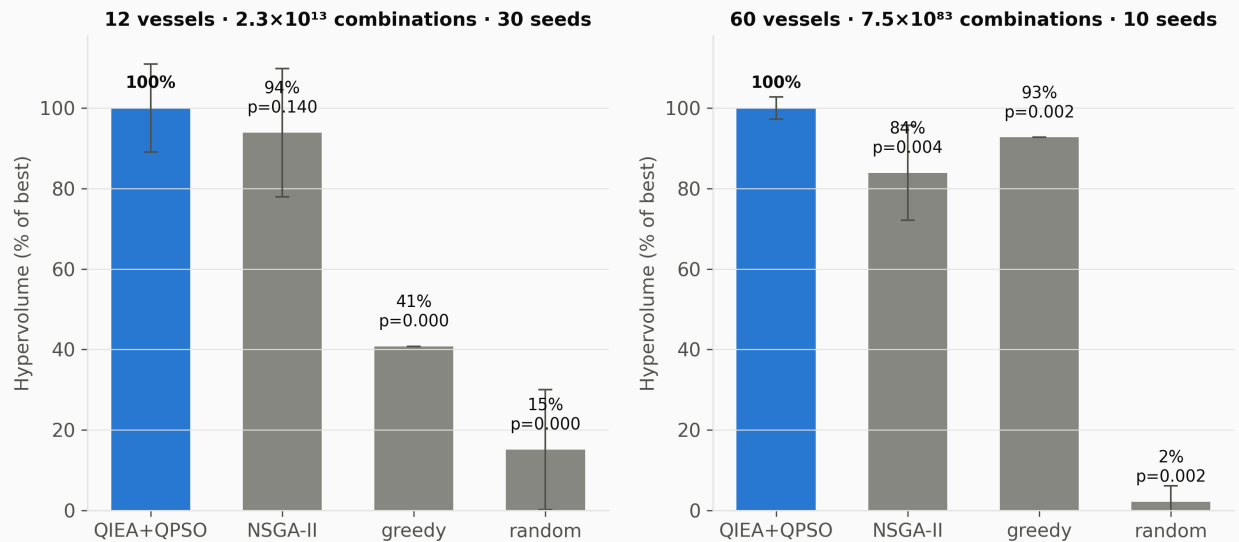
empirical interval coverage
against a nominal 80%

54 s

200 vessels, 10²⁷⁹ combinations
feasible plan found every run



Quantum-inspired optimizer vs conventional methods, equal evaluation budget (6,000)

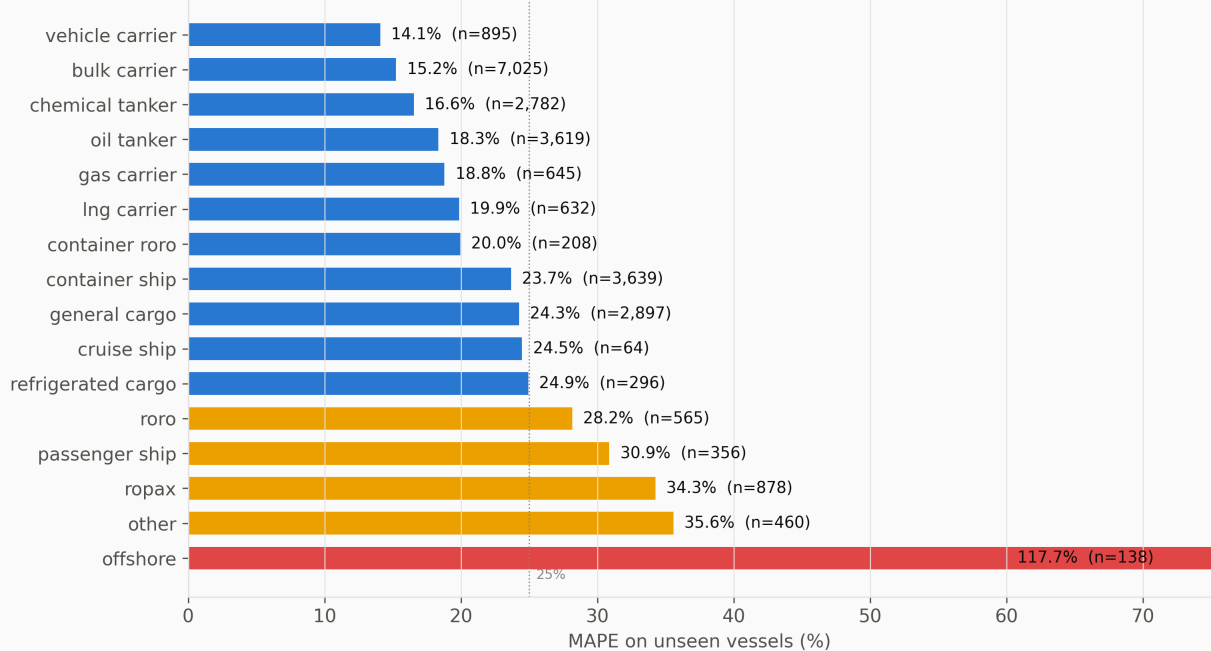


Shared reference point · Wilcoxon signed-rank paired by seed · large instance: QIEA better than every baseline, $p \leq 0.004$

Equal evaluation budget, shared reference point, Wilcoxon paired by seed. Tie at 12 vessels; QIEA better than every baseline at 60, $p \leq 0.004$.

SLIDES 3 · 4

Accuracy per vessel class - reported, not hidden behind the average

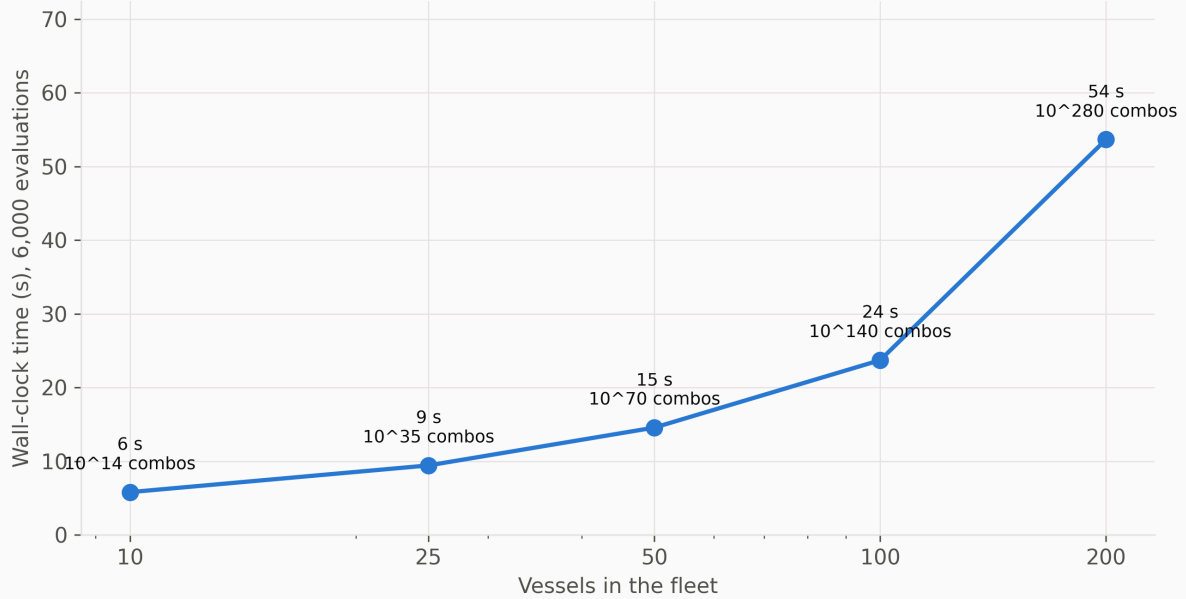


Offshore: 96% of rows lack a design certificate and are flagged cold-start with wide intervals (see chart 07).

Accuracy per class — the weak ones shown, not averaged away.

SLIDE 4

Runtime grows roughly linearly with fleet size; feasible plans found at every size



Feasible in 3/3 seeds at every size. Front thins above ~50 vessels at this budget (documented limitation).

Runtime grows roughly linearly with fleet size.

SLIDE 4

Viability — a real programme, a real first user

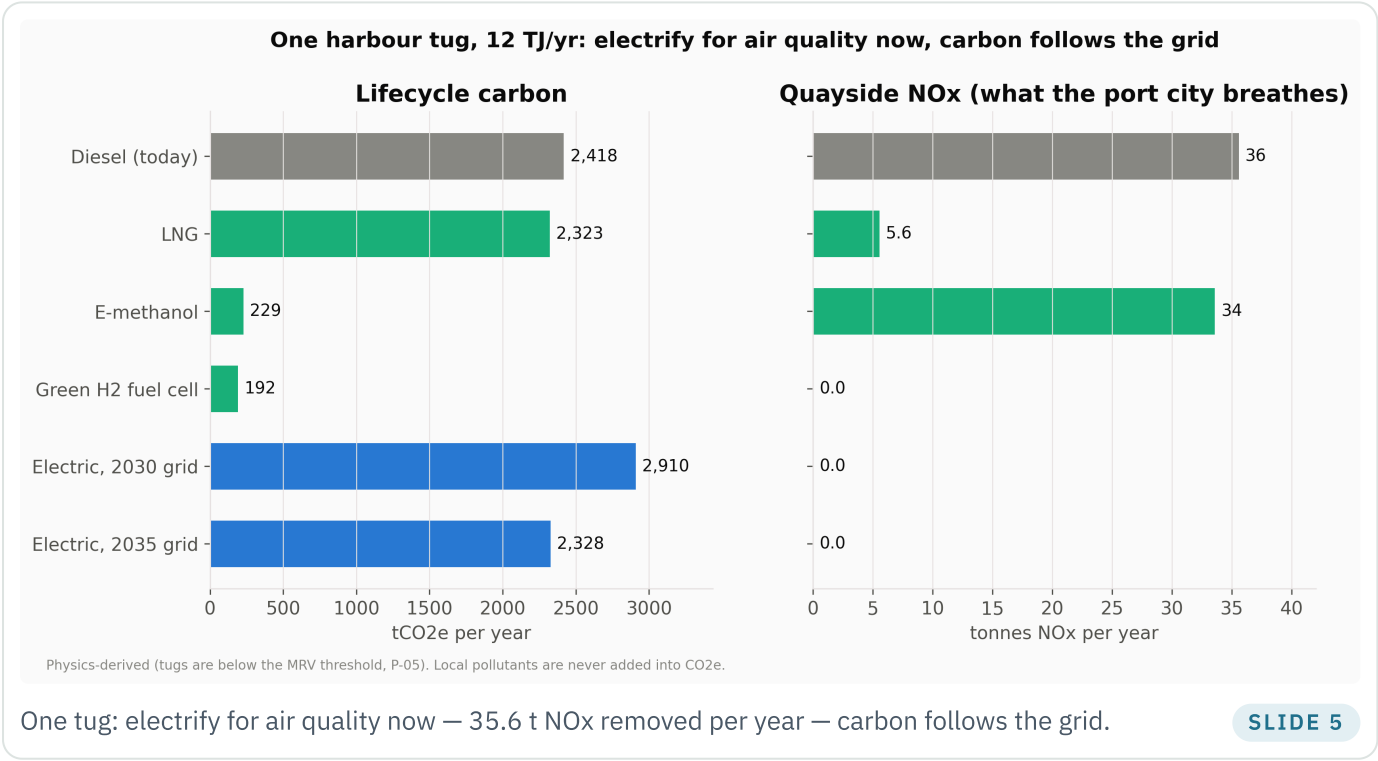
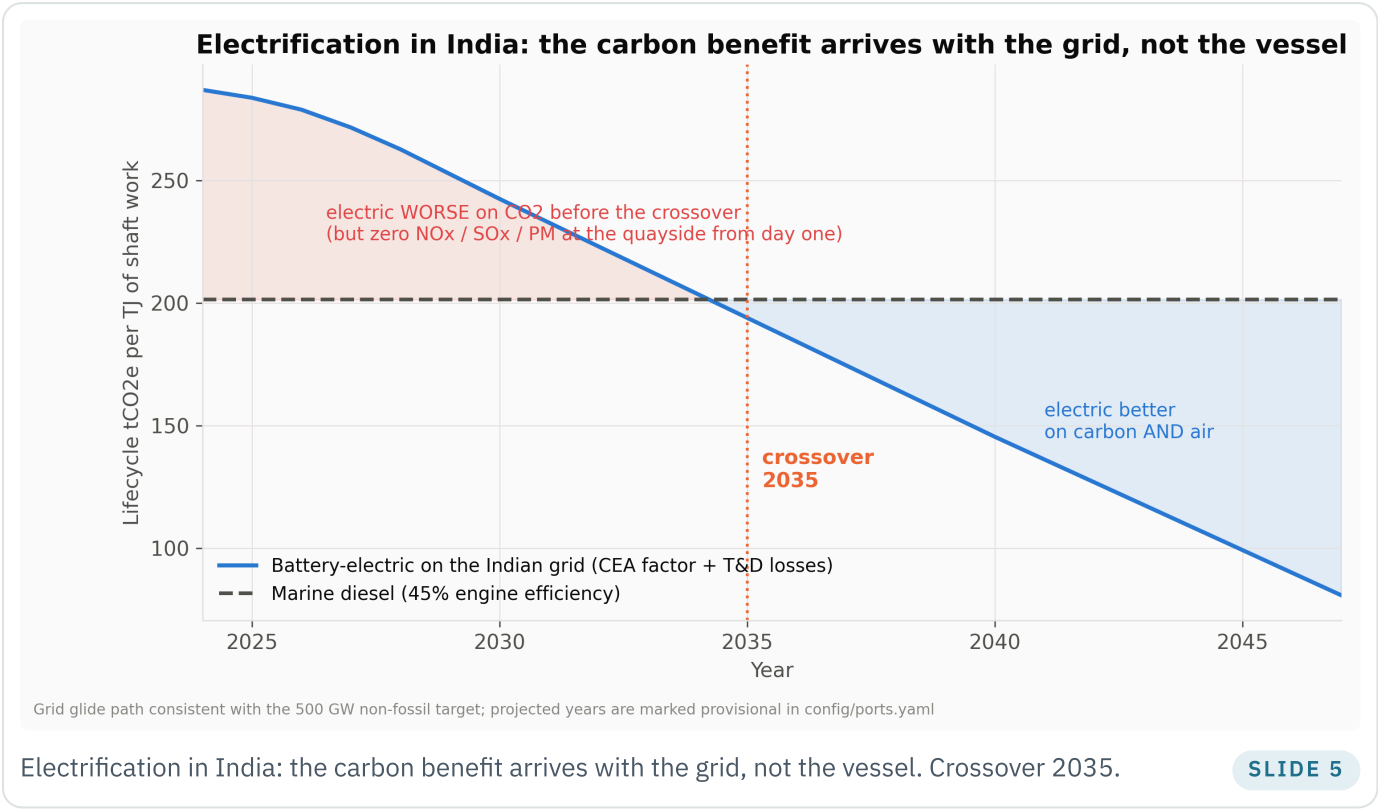
- **Green Tug Transition Programme.** Deendayal, JNPA, Visakhapatnam and V.O. Chidambaranar have placed electric-tug work orders. Each faces exactly our fleet-mix, fuel and duty-cycle decision.
- **Mormugao Port Authority** signed a strategic quantum-technology agreement with Egreen Quanta (July 2026) and runs Harit Shrey, India's first green-ship incentive. Our emission scores plug straight into it — the sponsor's live pilot.
- **Harit Sagar.** Ports must cut emissions per tonne of cargo 30% by 2030 and 70% by 2047. The KPI tracker reports the glide path and on/off-track status.
- **Business model.** Licence to port authorities / MoPSW, on-premises, annual support. Data stays in India.

Risks, each with a mitigation already built

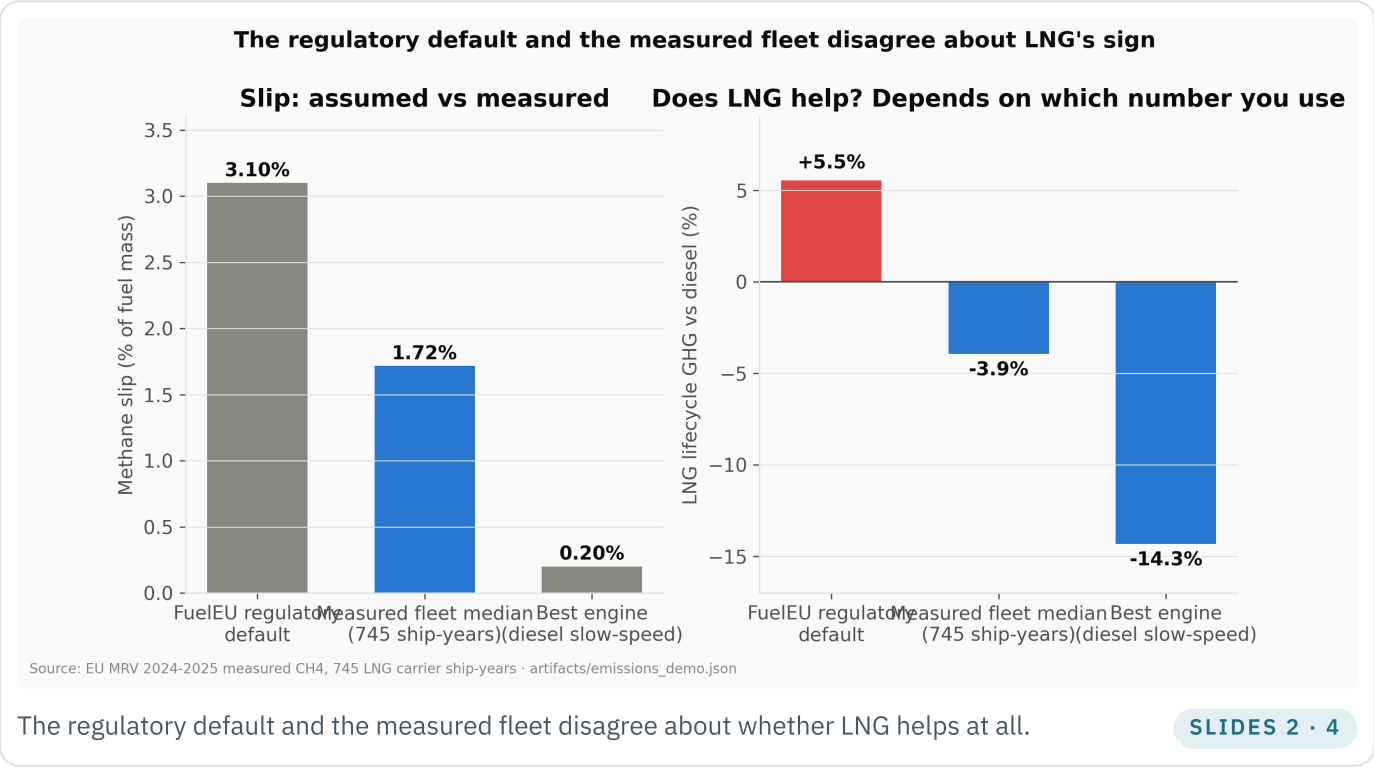
RISK	MITIGATION
Data scarcity for Indian coastal vessels and tugs — MRV has no ship under 5000 GT	Physics + vessel-class cold-start path: the model flags what it cannot see and widens intervals (cold rows: 84.2% coverage vs 80% nominal). Tug figures are labelled physics-derived.
"Quantum-inspired" is hype	Benchmarked at equal budget over 30 seeds. We publish the tie (12 vessels, $p = 0.14$) and the win (60 vessels, $p = 0.004$). Never claim quantum speedup.
Emission factors are uncertain	Every factor in config with a source and confidence tag. CH ₄ and N ₂ O now measured from 26,621 ship-years. Every result reports its weakest input's confidence.
Port fuel availability changes fast	Year-indexed config; "planned" deliberately does not count as available. Specific years are placeholders to confirm with port authorities — stated openly.
IMO Net-Zero Framework unsettled	Policy is a scenario toggle, not code. Harit Sagar and EU rules apply regardless.
Coal-heavy grid makes shore power look worse	Modelled honestly (crossover 2035) and paired with the immediate local air-quality benefit.
Live demo failure	Everything runs offline; pre-computed large scenario; recorded backup.

Say these before a judge finds them. Offshore vessels 117.7% MAPE (96% lack a design certificate; flagged cold-start). Ro-pax and passenger 30–34% (hotel-load-dominated). The Pareto front thins above ~50 vessels at a 6,000-evaluation budget. MILP verification is exact for the speed-discretised problem only. Nothing has run on quantum hardware.

Impact & benefits



IMPACT	NUMBER	SCENARIO
Avoids a wrong procurement	Stack-only scores grey ammonia at -100% ; the truth is +40% . A port buying on the wrong column would increase emissions and report reaching zero.	1 TJ shaft work, AR6 GWP100
Local air quality, immediately	Each electrified harbour tug removes 35.6 t NOx, 1.25 t SOx, 187 kg PM from the port city per year	12 TJ/yr tug, Mormugao
Carbon on the honest timeline	Electric tug CO ₂ : +20% vs diesel in 2030 → -4% in 2035 → -60% in 2047	CEA grid path
Fleet trade-off made visible	-91% GHG for +45% cost , with 14 feasible intermediate mixed-fuel plans	12 vessels, 3 routes, 2047
Slow steaming, correctly	14→12 kn saves 23.9%/mile on a bulk carrier, 0.1%/mile on a cruise ship	class hotel-load share
Harit Sagar tracked	35% clean-fuel share puts a 1 Mt port on track for -30% by 2030; 85% for -70% by 2047	tracker output
Range reality for hydrogen	Retrofit range 7% of diesel → tugs and ferries only; 4,842 t of cargo lost on a purpose-built 5,000 nmi tank	F-04 / F-05



Who benefits

BENEFICIARY	BENEFIT
Port authorities (GTTP)	A defensible fleet-transition plan with the CO ₂ and air-quality cases separated, infrastructure constraints respected
Mormugao Port	Emission scores that plug into Harit Shrey; the sponsor's own pilot site
MoPSW / Harit Sagar	Per-tonne-of-cargo intensity against the 2030 / 2047 targets, per port
Coastal & inland operators (Harit Nauka)	Which vessels, which fuels, which speeds — a Pareto front, not one number
National Quantum Mission	A real maritime problem already formulated as QUBO, ready for domestic hardware
Public health in port cities	Quantified NO _x / SO _x / PM removal, kept separate from carbon

Research & references

Data

- EU MRV "Publication of information", EMSA THETIS-MRV — mrv.emsa.europa.eu (2018–2025; pinned versions in `data/raw/mrv/manifest.json`)
- CEA CO₂ Baseline Database for the Indian Power Sector

Standards and regulation

- IMO MEPC.308(73) — 2018 EEDI Guidelines (LHV and C_f values)
- IMO Fourth GHG Study 2020 — our measured CH₄ reproduces its default exactly
- FuelEU Maritime Annex I/II — well-to-tank defaults, C_{slip} values
- IPCC AR5 / AR6 WG1 — GWP100; EMSA's use of AR5 recovered exactly from 28,125 filings
- Harit Sagar Green Port Guidelines; Green Tug Transition Programme — MoPSW
- PIB: Mormugao Port Authority – Egreen Quanta Strategic Quantum Technology agreement, July 2026
- IMO: Net-Zero Framework talks to resume in 2026

Methods

- Han & Kim (2002), *Quantum-inspired evolutionary algorithm for a class of combinatorial optimization*, IEEE TEC
- Sun, Feng & Xu (2004), *Particle swarm optimization with particles having quantum behavior*, IEEE CEC
- Deb et al. (2002), NSGA-II · Google OR-Tools CP-SAT
- ScienceDirect: *Improved quantum genetic algorithm for ship speed optimization*

Competitors reviewed

ZeroNorth voyage optimisation · Wärtsilä Fleet Optimisation Solution · DNV Alternative Fuels Insight · Maritime Executive on quantum port scheduling (Feb 2026)

Every number, with its source

Prediction model — artifacts/evaluation.json, unseen vessels

MODEL	MAE	RMSE	MAPE	MEDAPE	R ²	80% COV
class mean (naive)	1528.5	2780.8	32.6%	23.1%	0.354	80.3%
ridge (log)	1482.6	2724.9	31.5%	22.2%	0.379	80.4%
Physics + XGBoost	958.7	2178.7	19.9%	13.4%	0.603	80.0%
Physics + XGBoost, monotone	995.4	2220.0	20.6%	14.0%	0.588	80.1%

Train 75,423 rows / 18,788 ships → test 25,105 rows / 6,263 ships. Unseen years (2024–25): best 24.4%. Warm rows (n = 23,652): 19.3% MAPE, R² 0.697. Cold rows (n = 1,453): 41.8%, coverage 84.2%. Per class: vehicle carrier 14.1 · bulk 15.2 · chemical tanker 16.6 · oil tanker 18.3 · LNG carrier 19.9 · container 23.7 · ro-pax 34.3 · offshore 117.7 (cold). Quantum-inspired hyperparameter tuning: a tie (QPSO 20.679 vs random 20.728 vs untuned 20.875) — reported as such.

Emissions — artifacts/emissions_demo.json, per TJ shaft work, AR6 GWP100

FUEL / PATHWAY	STACK-ONLY TCO ₂ E	LIFECYCLE TCO ₂ E	STACK SAYS	TRUTH
MGO fossil	75.1	90.7	0%	0%
LNG fossil	57.3	87.1	−24%	−4%
Methanol grey	66.3	100.2	−12%	+10%
E-methanol	0.0	8.6	−100%	−91%
Ammonia grey	0.0	126.5	−100%	+40%
Ammonia blue	0.0	44.7	−100%	−51%
Ammonia green	0.0	18.6	−100%	−79%
Hydrogen grey	0.0	99.1	−100%	+9%
Hydrogen green	0.0	7.2	−100%	−92%

Methane slip: FuelEU default 3.10% → LNG +5.5% vs diesel; measured median 1.72% → −3.9%; best engine 0.20% → −14.3%. Measured spread p25 0.63%, p95 2.97% (745 ship-years). Conventional-fuel CH₄ 0.050 g/kg and N₂O 0.180 g/kg measured across 26,621 ship-years. EU GWP set recovered exactly: CH₄ 28.00, N₂O 265.00, residual 1.6×10^{−14}. Electric vs diesel (201.5): 2026 +38% · 2030 +20% · 2035 −4% · 2040 −28% · 2047 −60%. Harbour tug 12 TJ/yr: diesel 2,418 tCO₂e / 35,597 kg NO_x / 1,249 kg SO_x / 187 kg PM; electric 2030 2,910 / 0 / 0 / 0; electric 2035 2,328 / 0 / 0 / 0. Hydrogen retrofit range 374 nmi vs 5,000 for MGO (7%); cargo displaced on a purpose-built 5,000 nmi tank: 4,842 t.

Optimizer — artifacts/optimizer_benchmark.json

Q-08: MILP proven optimum INR 108,523,810 · QIEA+QPSO 30/30 exact, mean and worst gap 0.0000%.

INSTANCE	ALGORITHM	HV MEAN	HV STD	FRONT	TIME	FEASIBLE	VS QIEA
12 vessels 2.33×10^{13} · 30 seeds	QIEA+QPSO	4.195e21	4.59e20	6.4	2.87 s	30/30	—
	NSGA-II	3.938e21	6.67e20	6.2	2.25 s	30/30	tie · p = 0.14
	greedy	1.710e21	8.0e05	3.0	0.05 s	30/30	QIEA better · p < 0.001
	random	6.346e20	6.24e20	1.7	1.82 s	30/30	QIEA better · p < 0.001
60 vessels 7.52×10^{83} · 10 seeds	QIEA+QPSO	3.885e26	1.07e25	4.8	15.3 s	10/10	—
	greedy	3.602e26	7.2e10	2.0	1.0 s	10/10	QIEA better · p = 0.002
	NSGA-II	3.260e26	4.57e25	2.8	8.6 s	10/10	QIEA better · p = 0.004
	random	8.381e24	1.55e25	0.6	11.9 s	6/10	QIEA better · p = 0.002

Scalability (3 seeds, 6,000 evals): 10 vessels 5.8 s · 25 → 9.4 s · 50 → 14.6 s · 100 → 23.7 s · 200 (3.9×10^{279}) → 53.7 s, feasible 3/3 at every size. QUBO: ground state matches CP-SAT exactly; penalty weight neither too low nor too high. QIEA vs random on knapsack with a known optimum: 100% vs 63% at 4 bits, 100% vs 53% at 8 bits, 96% vs 26% at 12 bits.

Judge questions, with the number

"Is this actual quantum computing?"

No. Quantum-*inspired*: classical algorithms using superposition-style Q-bit encoding and rotation-gate updates, on ordinary hardware. The discrete core is also written as a QUBO, verified to match the exact optimum, so it can move to annealers or National Quantum Mission hardware when practical. Never say "quantum speedup".

"Why not just use a normal GA?"

We benchmarked exactly that. At 12 vessels it is a tie ($p = 0.14$). At 60 vessels the quantum-inspired optimizer beats NSGA-II with $p = 0.004$ and greedy with $p = 0.002$ at equal budget. The advantage appears when the space is large and rugged — chart 13 shows the ties too.

"Explain your encoding."

One Q-bit per binary decision; route and fuel by binary expansion, decoded modulo the option count so every string is valid; speed continuous via QPSO. Chart 14 and the Slide 3 text.

"What's the rotation angle and how did you choose it?"

$0.03\pi \cdot n_bits / n_generations$, calibrated by a documented sweep across three instance sizes. A fixed angle from a small problem lost 100× hypervolume on a large one — we show the failure.

"Have you run on a real quantum device?"

No. The QUBO's ground state matches CP-SAT on small instances, classically. That is the readiness claim, no more.

"How is this different from ZeroNorth / Wärtsilä?"

They optimise one voyage. We decide which ships, which fuels, which speeds across a fleet over years, with lifecycle emissions and Indian port constraints. Different decision level.

"Where does your data come from?"

EU MRV — 106,922 verified regulatory filings, 25,967 ships, 2018–2025, from EMSA's public API, version-pinned. Plus CEA grid data and IMO / FuelEU factors, each tagged with a confidence level. No synthetic rows.

"How accurate is your fuel prediction?"

19.9% MAPE on 6,263 ships never seen in training, R^2 0.603, leakage-safe split by vessel. Warm rows 19.3%; cold rows (41.8%) reported separately with 84% interval coverage.

"Isn't LNG a clean fuel?"

Only partly. Measured methane slip across 745 LNG carrier ship-years is 1.72% of fuel — with methane at 28–30× CO₂ that erases most of LNG's 24% combustion advantage, leaving –4%. The regulatory default (3.1%) would even make it worse than diesel.

"Is shore power actually green in India?"

On carbon, not until ~2035 — the grid is too coal-heavy for a 45%-efficient marine diesel to lose. On local air quality, immediately: 35.6 t NO_x per tug per year off the quayside. We show both.

"What about ships with no sensors?"

Physics-informed model with vessel-class baselines; the cold-start path flags them and widens the interval. Tugs are handled exactly this way.

"Who would actually use this?"

GTTP ports — Deendayal, JNPA, Visakhapatnam, V.O. Chidambaranar have placed electric-tug orders. First case: Mormugao, the sponsor's pilot site, plugging into Harit Shrey.

"How does this scale to 1,000 vessels?"

Runtime is roughly linear: 200 vessels in 54 s at 6,000 evaluations. Front density thins above 50 at that budget; more budget fixes it, and the cost is tabled.

What not to claim

Never "quantum" alone — always "quantum-inspired". No speedup, no hardware execution.

The 12-vessel benchmark is a tie ($p = 0.14$), not a win.

Port fuel-availability years are placeholders pending port-authority confirmation.

Tug figures are physics-derived; no tug is in MRV.

The 11% within-ship floor is a lower bound for any method, not a target for unseen vessels.

Never add NO_x / SO_x / PM into a CO₂e total.

Offshore and ro-pax are known weak classes; do not cite them as representative.

Every number must carry its scenario. "Up to X%" with no source loses the room.

Regenerate everything: `scripts/evaluate_model.py --calibrate-intervals` ·
`scripts/emissions_demo.py` · `scripts/benchmark_optimizer.py --seeds 30` ·
`scripts/make_charts.py` . Full text with sources in `docs/REPORT_FOR_PPT.md` .